Homework #3

CS5402 — Intro To Data Mining

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1 C4.5

Split	Entropy Less Than	Entropy Greater Than	Entropy Before Split	Entropy After Split	Information Gain
8	0	1.55459	1.579	1.435	0.14400
10	1	1.57262	1.579	1.48453	0.09447
15	0.9183	1.52193	1.579	1.38263	0.19637
20	0.81128	1.39215	1.579	1.21342	0.36558
24	0.97095	1.40564	1.579	1.23845	0.34055
28	0.9183	0.98523	1.579	0.95434	0.62466
28	1.37878	1	1.579	1.20396	0.37504
28	1.5	0.97095	1.579	1.29652	0.28248
29	1.53049	1	1.579	1.36726	0.21174
30	1.52193	0.9183	1.579	1.38263	0.19637
30	1.49492	0	1.579	1.26493	0.31407
31	1.55459	0	1.579	1.435	0.14400
33	1.57662	0	1.579	1.57662	0.00238

2 Grouping

Grouping	Entropy	Entropy Before Split	Entropy After Split	Information Gain
Coke-Pepsi	1.5219	1.565	1.2652	0.2998
Mountain Dew	1.5850	1.565		
Coke-Mountain Dew	1.4591	1.565	1.0943	0.4706
Pepsi	0	1.565		
Pepsi-Mountain Dew	1.3710	1.565	0.9623	0.6028
Coke	0.9183	1.565		

No, they should not be grouped; the respective information gains for the groupings are less than the information gain without grouping.

3 J48

For the deepest subtree, `credit_history`, we get the following error for the individual leaves:

$$U_{90\%}(0,\,1) + U_{90\%}(0,\,1) + 7 \times U_{90\%}(1,\,7) + 0 + 2 \times U_{90\%}(0,\,2) \approx$$
$$0.95 + 0.95 + 7 \times 0.5207 + 0 + 2 \times 0.77639 \approx 3.19709$$

If the subtree would just be replaced by a leaf **good (9/2)**, predicted error would be

$$9 \times U_{90\%}(2,\,9) \approx 9 \times 0.47009 \approx 4.23081$$

Therefore, the new tree would be

```
property_magnitude = car
|    personal_status = male div/sep: bad (1.0)
|    personal_status = female div/dep/mar: good (7.0)
|    personal_status = male single: good (9.0/2.0)
|    personal_status = male mar/wid: good (1.0)
|    personal_status = female single: good (0.0)
```

For this subtree, we get the following error for individual leaves:

$$U_{90\%}(0,\,1) + 7 \times U_{90\%}(0,\,7) + 9 \times U_{90\%}(2,\,9) + U_{90\%}(0,\,1) + 0 \approx$$
$$0.95 + 7 \times 0.348169 \times 0.470090 \approx 7.61793$$

Again, if the subtree would be replaced by a leaf **good (17/2)**, predicted error would be

$$17 \times U_{90\%}(2,\,17) \approx 17 \times 0.70420 \approx 5.0286$$

Therefore, the new tree would be

```
property_magnitude = car: good (17/2)
```

4 Missing Value

4.1 Outdoor

4.1.1 Hair = Short

$$\text{entropy} = -\frac{1 + {}^{2}\!/_{7}}{2 + {}^{2}\!/_{7}} \log_2 \frac{1 + {}^{2}\!/_{7}}{2 + {}^{2}\!/_{7}} - \frac{1}{2 + {}^{2}\!/_{7}} \log_2 \frac{1}{2 + {}^{2}\!/_{7}} = 0.9887$$

4.1.2 Hair = Long

$$\text{entropy} = \frac{0}{\vdots} \cdots = 0$$

4.2 Indoor

4.2.1 Hair = Short

$$\text{entropy} = -\frac{1}{1+{}^{2}\!/_{7}}\log_2\frac{1}{1+{}^{2}\!/_{7}} - \frac{{}^{2}\!/_{7}}{1+{}^{2}\!/_{7}}\log_2\frac{{}^{2}\!/_{7}}{2+{}^{2}\!/_{7}} = 0.7642$$

4.2.2 Hair = Long

$$\text{entropy} = -\frac{1}{1}\log_2\frac{1}{1} = 0$$

4.3 Both

4.3.1 Hair = Short

$$\text{entropy} = -\frac{1+{}^{3}\!/_{7}}{1+{}^{3}\!/_{7}}\log_2\frac{1+{}^{3}\!/_{7}}{2+{}^{3}\!/_{7}} = 0 = 0$$

4.3.2 Hair = Long

$$\text{entropy} = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$$